

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE CODE:** MLA03

**COURSE NAME:** Reinforcement learning for Environment and Agent

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### **COURSE OUTCOME COVERED**

**CO2:** *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. ( BL2, BL3)*

*Assessment Tool Weightage: 30%*

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Dr DUMPALA PRASANTH

**QUESTIONS: 10**

**Total Marks: 50**

### **Annexure A** **Application - Based Questions**

**Duration: 2Hrs**

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. **(5 Marks)**
2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. **(5 Marks)**
3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time. **(5 Marks)**
4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. **(5 Marks)**
5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism. **(5 Marks)**

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. **(5 Marks)**

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization. **(5 Marks)**

8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency. **(5 Marks)**

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance. **(5 Marks)**

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

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***Code of Conduct:***

*I B. Lokesh (192472339) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

B. Lokesh

***Signature of the student:***

## RUBRICS FOR CALCULATION

| Criteria                         | Weightage | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

### **1. RL Framework for Ride-Sharing Driver–Passenger Matching**

A ride-sharing company can use Reinforcement Learning (RL) to match passengers with suitable drivers and reduce waiting time.

- State Space: Current locations of drivers and passengers, number of available drivers, passenger demand, traffic conditions, and estimated travel time.
- Action Space: Assign a particular driver to a passenger, keep the driver available, or redirect the driver to a high-demand area.
- Reward Function: Give positive rewards for quick and successful matching and negative rewards for long passenger waiting time, driver idle time, cancellations, and long travel distances.
- Policy: The RL agent learns the best matching strategy for different traffic and demand conditions.
- Goal: Over time, the system learns to minimize waiting time, reduce idle drivers, and improve overall service quality.

### **2. RL for Personalized E-Learning Recommendations**

An e-learning platform can use RL to recommend study materials according to each student's learning behavior.

- State: Student's current knowledge level, previous materials viewed, quiz scores, time spent studying, and learning progress.
- Actions: Recommend familiar/easy content, advanced content, videos, quizzes, or new topics.
- Exploitation: Recommends materials that have already been successful for the student based on previous performance.
- Exploration: Recommends new or less-tested materials to discover potentially better learning resources.
- Balance: A suitable strategy such as  $\epsilon$ -greedy can be used, where most recommendations exploit known good choices while a small percentage explore new content.

Thus, RL provides personalized recommendations while continuously improving the student's learning experience.

### **3. RL Framework for a Robotic Vacuum Cleaner**

A robotic vacuum cleaner can use RL to learn an efficient cleaning path while avoiding obstacles.

- State: Robot's current position, battery level, dirt level, nearby obstacles, and already-cleaned areas.
- Actions: Move forward, backward, left, right, turn, clean an area, or return to the charging station.
- Policy: The policy determines the best action for each state. For example, if an obstacle is detected, the policy can select a safe alternative direction.
- Value Function: It estimates the future reward of being in a particular state. States leading to more cleaning with less battery consumption receive higher values.
- Learning: Through repeated cleaning operations, the robot updates its value estimates and policy, resulting in better coverage, fewer collisions, and lower energy consumption.

#### **4. RL-Based Cryptocurrency Trading System**

A cryptocurrency trading system can use RL to decide whether to Buy, Sell, or Hold.

- State: Cryptocurrency price, trading volume, price changes, market trends, portfolio balance, and technical indicators.
- Actions: Buy, sell, or hold a cryptocurrency.
- Reward: Reward can be based on profit or portfolio value, while losses can receive negative rewards.
- Reward Design Challenges: Cryptocurrency markets are highly volatile, and short-term profit may encourage risky trading. Transaction costs, market fluctuations, and delayed returns must also be considered.
- Effect of Poor Reward Design: If the reward only considers immediate profit, the agent may take excessive risks or perform frequent trades. Therefore, the reward should consider profit, risk, transaction costs, and long-term performance.

#### **5. RL Framework for Smart Grid Electricity Distribution**

RL can be used in a smart grid to distribute electricity efficiently according to demand and available resources.

- State Space: Current electricity demand, power generation, battery level, renewable energy availability, grid load, and electricity price.

- Action Space: Increase or decrease power distribution, charge/discharge batteries, use renewable sources, or shift loads.
- Reward Function: Positive reward for satisfying electricity demand efficiently and using renewable energy; negative reward for power loss, overload, high cost, and unmet demand.
- Policy: The RL agent learns the best distribution decision for different grid conditions.
- Goal: Reduce energy wastage and operating cost while maintaining reliable electricity supply.

## **6. RL for Healthcare Medication Dosage**

An RL-based healthcare monitoring system can learn appropriate medication dosage decisions based on a patient's changing condition. In real healthcare, such a system should be used only with appropriate clinical supervision.

- State: Patient's vital signs, symptoms, laboratory results, previous dosage, medication response, and health condition.
- Actions: Select an appropriate dosage level, maintain the current dosage, or adjust treatment.
- Reward: Positive reward for maintaining the patient within desired health ranges and negative reward for adverse effects or poor health outcomes.
- Policy: The policy maps the patient's current state to an appropriate treatment decision.
- Learning: By observing patient responses over time, the system can improve its treatment policy and aim for better outcomes while minimizing adverse effects.

## **7. RL-Based Drone Delivery Navigation**

A drone delivery system can use RL to find efficient and safe routes between the warehouse and delivery location.

- State: Drone's current location, battery level, altitude, weather conditions, obstacles, traffic restrictions, and destination.
- Actions: Move forward, backward, left, right, change altitude, or select another route.
- Exploration: The drone tries different routes to discover shorter or safer paths.
- Exploitation: It uses routes that have already produced good results, such as low travel time and energy consumption.

- Balance: An  $\epsilon$ -greedy strategy can be used to occasionally explore new routes while mostly selecting the best-known route.

This allows the drone to continuously improve route efficiency while considering safety and battery usage.

## **8. RL for Manufacturing Robot Assembly**

A manufacturing robot can use RL to optimize assembly operations and reduce production time.

- State: Robot position, component position, assembly stage, tool status, and remaining production tasks.
- Actions: Pick, place, rotate, assemble components, change tools, or move to another position.
- Policy: The policy determines which assembly action should be performed in each state.
- Value Function: It estimates the expected future reward of an assembly state. States that lead to faster and successful assembly receive higher values.
- Improvement: By repeatedly performing assembly tasks, the robot updates its policy and value function. This can reduce errors, assembly time, and energy consumption while increasing productivity.

## **9. RL for Online Advertising and Click-Through Rate**

An online advertising platform can use RL to select advertisements that are likely to receive user clicks.

- State: User interests, browsing behavior, previous clicks, time, device type, and current advertisement information.
- Actions: Select and display one advertisement from available advertisements.
- Reward: A positive reward can be given for a click or conversion, while a negative/low reward can be assigned for ignoring an advertisement.
- Reward Design Challenges: Optimizing only clicks may encourage misleading or clickbait advertisements. It may also ignore purchases, user satisfaction, long-term engagement, and advertising cost.
- Impact: Poor reward design can increase short-term CTR but reduce user trust and long-term business performance. A better reward should consider clicks, conversions, relevance, and long-term user engagement.

## **10. RL Framework for Smart Irrigation**

A smart irrigation system can use RL to determine when and how much water should be supplied to crops.

- **State Space:** Soil moisture, temperature, humidity, rainfall prediction, crop growth stage, weather conditions, and water availability.
- **Action Space:** Irrigate with a small, medium, or large amount of water, or do not irrigate.
- **Reward Function:** Positive reward for maintaining suitable soil moisture and healthy crop growth. Negative reward for excessive water usage, insufficient moisture, or unnecessary irrigation.
- **Policy:** The RL agent learns the best irrigation action for different soil and weather conditions.
- **Goal:** Reduce water consumption while maintaining healthy crop growth and improving irrigation efficiency.